

OFFLINE VOICE & TEXT AI ASSISTANT

A privacy-first edge AI assistant for voice and text interaction

HACKATHON PROJECT BRIEF

Built with Python, Whisper speech-to-text, and a deployment path for Qualcomm AI Hub / Snapdragon NPU optimization.

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Repository: github.com/sriyanshmishra/ai-voice-assistent

1. Executive Summary

Offline Voice & Text AI Assistant is a lightweight edge-AI prototype that accepts either microphone audio or typed text. Voice input is transcribed locally with OpenAI Whisper Tiny, then passed to a simple intent-handler layer. The design keeps audio and text processing on-device after the model is cached, reducing dependency on cloud services and improving privacy for everyday interactions.

The project is intentionally structured for a Snapdragon deployment path: the current development fallback uses Hugging Face Whisper, while the same input contract can be connected to a Qualcomm AI Hub-exported model for NPU execution.

2. Problem

	Voice assistants commonly send recordings to remote servers, creating latency, connectivity, and privacy concerns.
	Users need a quick way to switch between voice and text depending on environment, accessibility, or hardware availability.
	Modern Snapdragon devices can run AI locally, but prototypes need a clear bridge from model experimentation to device-optimized inference.

3. Solution

The assistant provides a compact, local-first interaction loop:

1	Choose voice or text input from the terminal interface.
2	Record a short microphone sample or accept typed text.
3	Transcribe voice locally with Whisper when voice mode is selected.
4	Route the resulting text to a replaceable intent-handler function.
5	Run from a cached model with WHISPER_LOCAL_ONLY=1 when offline operation is required.

4. Technical Architecture

Input layer	sounddevice microphone capture or terminal text input
Pre-processing	16 kHz float32 audio passed to the Whisper processor
Inference	Whisper Tiny local model with PyTorch inference mode; designed for replacement with a Qualcomm AI Hub optimized runtime
Application layer	Transcribed text flows into <code>process_text_input()</code> , a clear extension point for commands and intents
Configuration	Environment variables control model selection and local-only loading

5. What Works Today

- Voice recording from the default microphone with a configurable five-second duration.
- Whisper-based transcription and local text decoding.
- Direct text input for testing without microphone access.
- Lazy model loading, so text mode can be tested before downloading Whisper.
- Offline-only mode after the model has been downloaded and cached.

6. Snapdragon / Qualcomm AI Hub Roadmap

Current prototype	Whisper Tiny via Hugging Face + PyTorch development fallback
Optimization	Export and compile the compatible Whisper graph with Qualcomm AI Hub for a target Snapdragon device
Runtime	Replace the model invocation behind the existing transcription interface with the device runtime
Productization	Add intents, wake-word handling, streaming audio, UI, and on-device evaluation metrics

7. Expected Impact

The project demonstrates a practical path to private, low-latency AI interaction on edge hardware. It can support environments with weak connectivity, reduce the need to transmit sensitive audio, and provide a modular foundation for accessibility tools, field assistants, education, and device control.

8. Repository

github.com/sriyanshmishra/ai-voice-assistant

Note: The current repository is a working prototype. Qualcomm NPU acceleration is a planned integration path and should be validated on the target Snapdragon hardware before making performance claims.